

Rain Zimin Yang

rainziminyang@gmail.com | +86 188 1152 8937 | [linkedin.com/in/ryazimn](https://www.linkedin.com/in/ryazimn) | ryazimn.github.io

EDUCATION

Columbia University Ph.D in Computer Science.	Starting in Sept 2026
University of British Columbia BSc Combined Honours in Computer Science and Mathematics with Distinction.	Sept 2023 - May 2026
- GPA: 92.8% - Undergraduate courses: Theory of Computation (CPSC 421), Randomized Algorithms (CPSC 436R), Quantum Computation (CPSC 436Q), Group Theory (MATH 322), Number Theory (MATH 437/538), Measure Theory (MATH 420/507), Probability (MATH 418/544), Fields and Galois Theory (MATH 422/501) - Graduate courses: Commutative Algebra (MATH 423/502), Algebraic Topology (MATH 427/527), Algebraic Geometry (MATH 532), Algebraic Number Theory (MATH 538), Communication Complexity (CPSC 536Z, audit)	

RESEARCH

Stabilizer Rank UBC Computer Science Supervisor: Daochen Wang	May 2026 - Now
Rational Degree Conjecture UBC Computer Science Supervisor: Daochen Wang	Apr 2025 - Apr 2026
- We proved that a natural complexity measure, the rational degree, of total Boolean functions is polynomially related to its Fourier degree, resolving the Rational Degree Conjecture posed in 1994. - Accepted to the 67th Annual Symposium on Foundations of Computer Science (FOCS 2026).	
Boundary Regularity for Solutions to the Plateau Problem UBC Mathematics Supervisor: Ailana Fraser	May - Aug 2024
- Investigated the existence of stable minimal surfaces with finite total curvature and straight-line boundary whose limit is the half-helicoid using tools from complex analysis. - Developed MATLAB visualizations to illustrate surfaces using Weierstrass-Enneper representation. - Presented in the UBC USRA Seminar on June 20, 2024. - Research supported via NSERC WLIURA.	

ACADEMIC COMPETITIONS

Honors in ICPC World Finals 2025 1st place in Canada, 8th place in North America (Team <i>Forgetful Functors</i>)	Sept 2025
11th place in ICPC NAC 2026 (Team <i>Diamond Dust</i>)	Mar 2026
ICPC Pacific Northwest Regional Contest 2025 - 2026 Bronze Medalist, Canadian Site Champion (Team <i>Diamond Dust</i>)	Nov 2025
ICPC Pacific Northwest Regional Contest 2024 - 2025 Silver Medalist, Canadian Site Champion (Team <i>Forgetful Functors</i>)	Nov 2024
William Lowell Putnam Mathematical Competition 2025 Top 431 out of 4329 Participant (90th percentile of competitors)	Dec 2025

TECHNICAL SKILLS

Programming Languages: C++, Java, Python

Mathematical Tools: MATLAB, Mathematica

Other: $\text{\LaTeX}$, Git

AWARDS AND SCHOLARSHIPS

- **Rick Sample Memorial Award** (\$2500), *UBC Computer Science* 2026
- **D F MacKenzie Scholarship** (\$1100), *UBC Faculty of Science* 2026
- **NSERC Undergraduate Student Research Award**, *UBC Computer Science* 2024, 2025, 2026
- **Stanley M Grant Scholarship in Mathematics** (\$4500), *UBC Mathematics* 2025
- **Dean of Science Scholarship** (\$210), *UBC Faculty of Science* 2025
- **Reginald Palliser-Wilson Scholarship** (\$4400), *UBC Mathematics* 2024
- **Entrance Scholarship in Mathematics** (\$10,424.12), *UBC Mathematics* 2023
- **Outstanding International Student Scholarship** (\$13,000), *UBC Faculty of Science* 2023
- **COMC Entrance Scholarship** (\$1000), *UBC Mathematics* 2023

TEACHING EXPERIENCE

- Teaching Assistant**, *UBC Computer Science* Jan - Apr 2026
Course: CPSC 420 Advanced Algorithm Design and Analysis | 71 students
- Teaching Assistant**, *UBC Computer Science* Sept - Dec 2025
Course: CPSC 421 Introduction to Theory of Computing | 53 students
- Teaching Assistant**, *UBC Computer Science* July - Aug 2025
Course: CPSC 320 Intermediate Algorithm Design and Analysis | 154 students
- Teaching Assistant**, *UBC Computer Science* Sept - Dec 2024
Course: CPSC 320 Intermediate Algorithm Design and Analysis | 386 students